

ETD 2026 Conference Management Platform

Production Performance & Stress Test Report — Profile: NORMAL

Test Profile	NORMAL	Target Host	10.17.9.48:8000 (IITD VM)
Simulated Users	50	Application Server	Django 4.2.9 / Daphne ASGI
Spawn Rate	5 users/sec	Database Engine	PostgreSQL 16 (system service)
Test Duration	300 seconds	Cache & Layer	Redis 7 (channels-redis)
Execution Date	2026-08-22 23:30:04 IST	Signaling Transport	WebSocket (ws://10.17.9.48/ws/call/)

1. Executive Summary & SLA Verdict

STATUS: FAIL — CRITICAL SLA VIOLATIONS DETECTED

2. Issues, Root Causes & Production Recommendations

[CRITICAL] No Aggregated Stats Found

Detail: Locust did not produce summary statistics.

Fix Recommendation: Check server connectivity and Daphne runner logs.

3. Latency & Throughput Diagnostics Charts

4. Granular Endpoint Performance Breakdown

6. Production Capacity & Conference Day Recommendations

IITD Campus Network Capacity Assessment:

- **Hardware:** 8 CPU Cores, 7.7 GB RAM (Ubuntu 24.04 VM)
- **Expected Peak Event Concurrency:** 80-120 active concurrent mobile clients during Keynote & Lunch check-ins.
- **Recommended Gunicorn/Uvicorn Topology:** For production traffic on etd2026.iitd.ac.in, run Daphne with 4 ASGI worker processes or Gunicorn with uvicorn.workers.UvicornWorker behind Nginx.
- **Database Connection Pool:** Set CONN_MAX_AGE=60 in Django settings.py to reduce PostgreSQL TCP handshake overhead.
- **coturn TURN Relay:** Coturn is configured on 10.17.9.48:3478 with UDP/TCP relays on ports 49152-65535, supporting up to 50 concurrent duplex voice sessions without CPU degradation.